higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

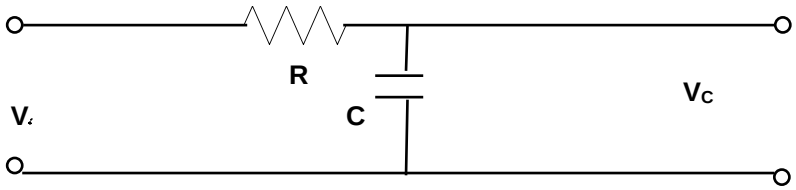
4 April 2024

This marking guideline consists of 9 pages.

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QUESTION 1: ALTERNATING CURRENT THEORY

1.1



Low-pass filter

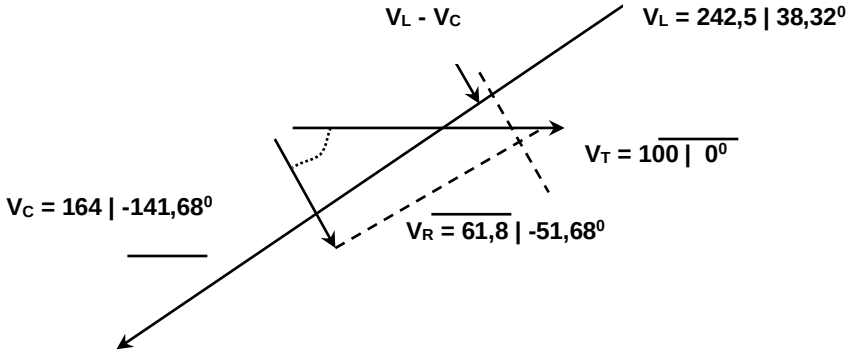
(One mark each for the resistor and capacitor, half mark each for the input and output) (3)

1.2 1.2.1 Greater

1.2.2 Square waveforms

(2 × 1) (2)

1.3 1.3.1



✓✓✓ (3)

1.3.2 **A**

$$X_L = \frac{V_L}{I_T}$$

$$= \frac{242,5 \mid 38,32^0}{5,15 \mid -51,68^0} \checkmark$$

$$= 47,087 \mid 90^0 \quad \Omega\checkmark$$

B $X_C = \frac{V_C}{I_T}$

$$= \frac{164 \angle -141{,}68^0}{5{,}15 \angle -51{,}68^0} \checkmark$$

$$= 31{,}845 \angle -90^0\ \Omega\checkmark$$

C $R = \frac{V_R}{I_T}$

$$= \frac{61{,}8 \angle -51{,}68^0}{5{,}15 \angle -51{,}68^0} \checkmark$$

$$= 12 \angle 0^0\ \ \Omega\checkmark$$

(3 × 2) (6)
[14]

QUESTION 2: POWER SUPPLIES

2.1 $V'_{dc} = V_{dc} - I_{dc}R_1$

$= 18 - 50 \text{ x } 10^{-3} \text{ x } 20\checkmark$

$= 17 \text{ V}\checkmark$ (2)

2.2 It blocks AC signals and passes DC signals. (2)

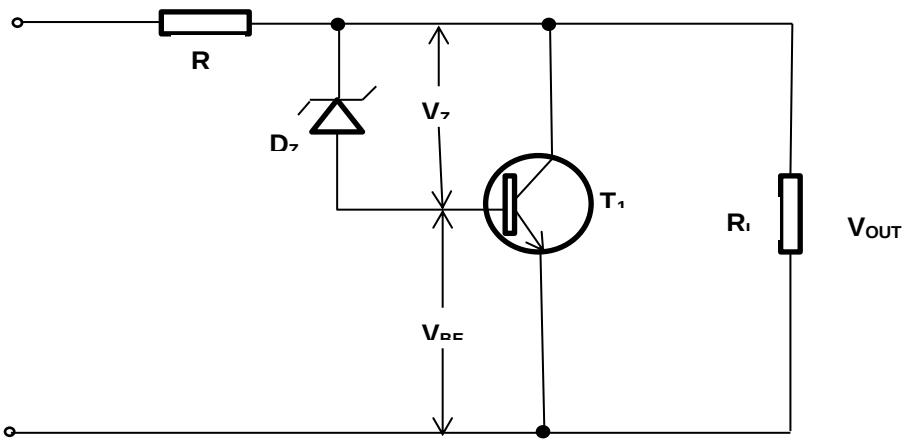
2.3 25 V = No-load voltage

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \text{ x } 100$$

$0{,}08 \text{x} V_{FL} = 25 - V_{FL}\checkmark\checkmark$

$V_{FL} = 23{,}15 \text{ V}\checkmark$ (3)

2.4



Half a mark for each correct component and a full mark for correctness of the sketch.

How the supply voltage is regulated:

- The value of the Zener diode (D_Z) is selected in such a way that the bias (V_{be}) of the transistor (T_1) is low enough to keep the transistor in an unsaturated condition.✓✓
- If the supply voltage increases, the Zener voltage (V_Z) will remain constant, but the base bias (V_{be}) will increase.✓✓
- This causes the transistor to be driven to greater saturation in order to let the collector voltage drop.✓✓
- This fall compensates for the increase in supply voltage and in this way the output voltage is stabilised.✓ (7)

[14]

QUESTION 3: TRANSISTOR AMPLIFIER

3.1 3.1.1 Supplies voltage to the junctions of the transistors.
3.1.2 Blocks DC signals whereas AC signals are passed.
(2 × 2) (4)

3.2 3.2.1

$$R_c = \frac{V_{CC} - V_{ce}}{I_C}$$

$$V_{ce} = V_{CC} + R_c I_c \therefore ✓$$

$$= 20 + 1000 \times 14 \times 10^{-3} ✓$$

$$= 34\ V ✓$$ (3)

3.2.2

$$R_b = \frac{V_{CC} - V_{be}}{I_b}$$

$$\therefore I_b = \frac{V_{CC} - V_{be}}{R_b} ✓$$

$$= \frac{20 - 0{,}7}{800 \times 10^3} ✓$$

$$= 24{,}125\ \mu ✓$$ (3)

3.2.3 $$\beta = \frac{I_C}{I_B}$$

$$= \frac{14 \text{ x } 10^{-3}}{24{,}125 \text{ x } 10^{-6}} \checkmark$$

$$= 580{,}311 \quad (2)$$

3.3 $$Z_2 = \frac{1}{h_{oe} - \frac{h_{oe} h_{re}}{h_{ie} + R_s}}$$

$$= \frac{1}{20\ x\ 10^{-6} - \frac{60 \text{ x } 2\ x\ 10^{-4}}{1200 + 0}} \checkmark$$

$$= 100 \text{ k}\Omega \checkmark$$

$$Z_O = \frac{R_C \text{ x } Z_2}{R_C + Z_2}$$

$$Z_O\ (R_C + Z_2) = R_C \text{ x } Z_2$$

$$1960\ (R_C + 100000) = R_C \text{ x } 100000$$

$$1960R_C + 196000000 = 100000R_C \checkmark$$

$$R_C\ (100000 + 1960) = 196000000$$

$$R_C = \frac{196000000}{98040}$$

$$R_C = 1999{,}184\ \Omega \checkmark$$

$$Z_i = h_{ie} - \frac{h_{fe}\, h_{fe} Z_L}{1 + h_{oe} Z_L}$$

$$Z_i = 1200 - \frac{60 \text{ x } 2\ \text{ x } 10^{-4} \text{x } 1999{,}184}{1 + (20 \text{ x } 10^{-6} \text{ x } 1999{,}184)} \checkmark$$

$$= 1176{,}932 \text{ k}\Omega \checkmark$$

$$Z_i = \frac{R_b \text{ x } Z_1}{R_b + Z_1}$$

$$Z_i\ (R_b + Z_1) = R_b \text{ x } Z_1$$

$$1169\ (R_b + 1176{,}932) = R_b \text{ x } 1176{,}932$$

$$1169R_b + 1375833{,}66 = R_b \text{ x } 1176{,}932$$

$$R_b\ (1176{,}932 + 1169) = 1375833{,}66 \checkmark$$

$$R_b = \frac{1375833{,}66}{7{,}932}$$

$$= 173{,}5 \text{ k}\Omega \checkmark \quad (6)$$

[18]

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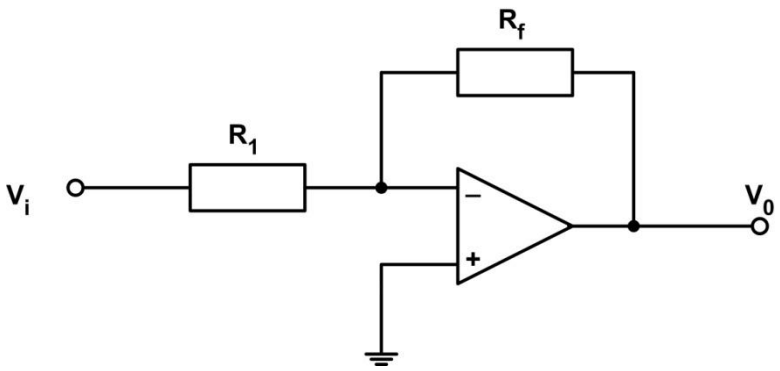
QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 4.1.1 The necessary difference between bias voltages at the input to obtain zero output voltages.

4.1.2 Input terminal of a differential amplifier, which produces an anti-phase output when a signal is applied.

(2 × 2) (4)

4.2 4.2.1



(ONE mark for each component) (3)

4.2.2 $V_O = \frac{-R_f}{R_1} \times V_{in}$

$R_f = \frac{V_o \times R_1}{V_{in}}$ ✓

$= \frac{20 \times 15000}{2}$ ✓

$= 150\ k\Omega$ ✓ (3)

[10]

QUESTION 5: INTEGRATED CIRCUITS

5.1 All unutilised inputs must be connected to V_{DD} or V_{SS} (2)

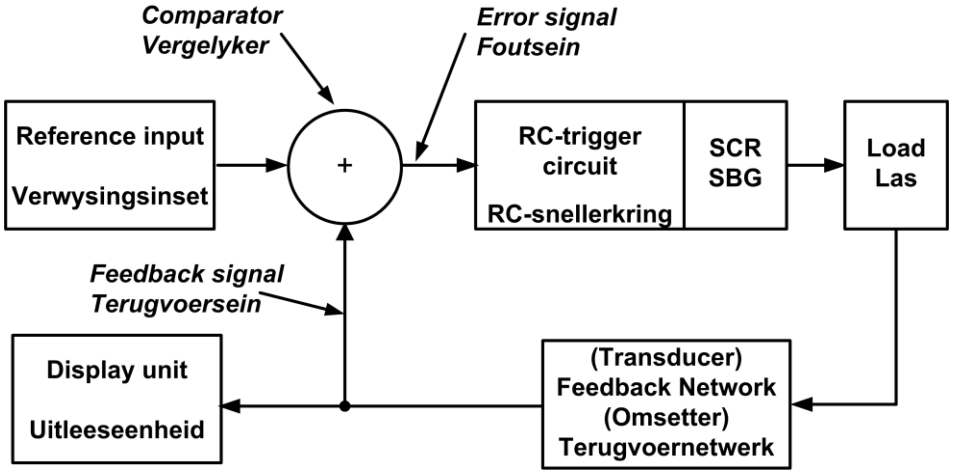
5.2 5.2.1 False
5.2.2 False
5.2.3 False

(3 × 1) (3)

[5]

QUESTION 6: ELECTRONIC PHASE CONTROL

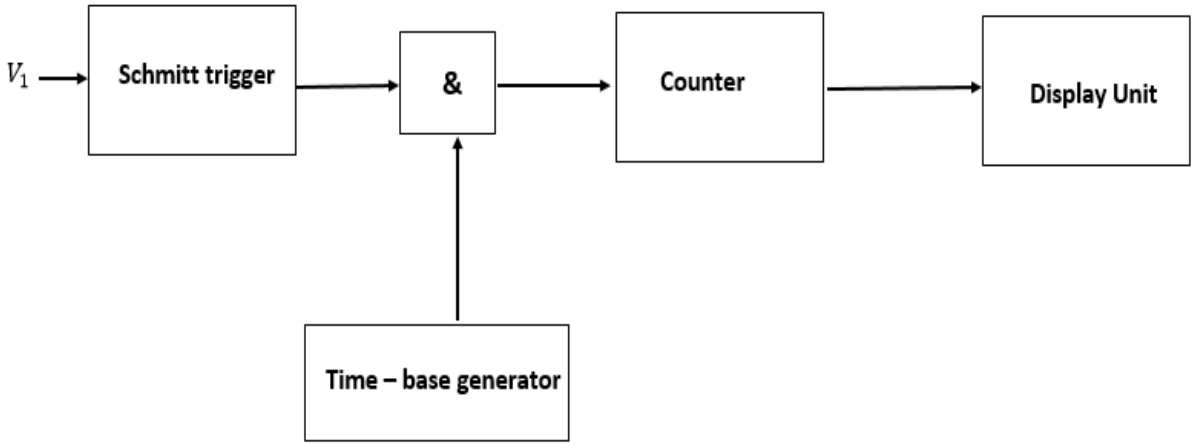
ELECTRONIC PHASE CONTROL



The comparator compares the feedback signal from the load with the reference signal.✓
As soon as the load deviates, an error signal appears on the output of the comparator,✓
which is used by the control circuit in order to bring the load condition back to normal.✓

(ONE for each correct block labelled) **[8]**

QUESTION 7: TEST EQUIPMENT



(ONE for each correct block label) **[5]**

QUESTION 8: OSCILLATORS

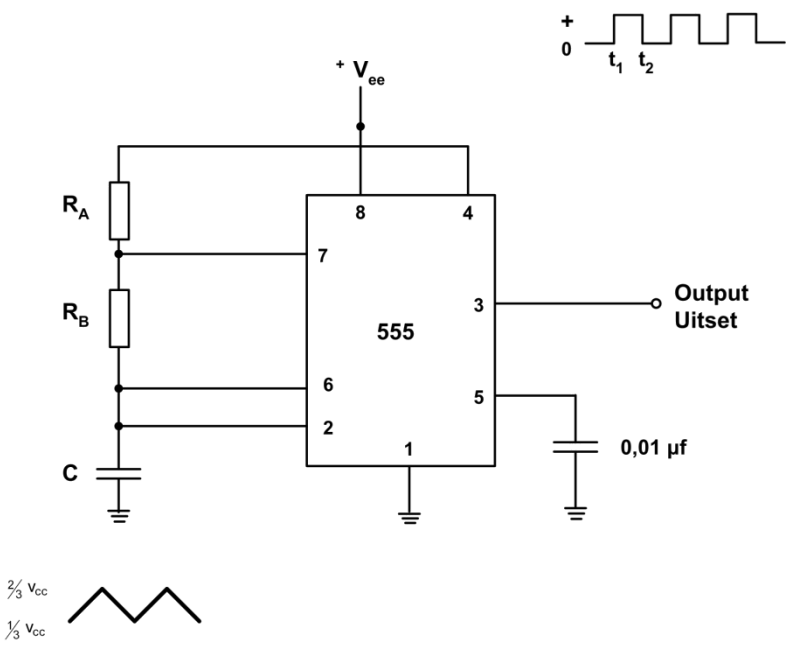
8.1 $$f_r = \frac{1}{2\pi\sqrt{L_T \cdot C}}$$

$$\therefore C = \frac{1}{(2 \cdot \pi \cdot f_r)^2 \cdot L_T} \checkmark$$

$$= \frac{1}{(2\times\pi\times300\times10^3)^2 \cdot 10\times10^{-3}} \checkmark$$

$$= 28.145 \text{ pf} \checkmark$$ (3)

8.2



(½ mark for each correct label for a total of 4 marks) (4)

8.3 $$f_o = \frac{1{,}443}{(R_A + 2R_B) \cdot C}$$

$$= \frac{1{,}443}{\left[90000 + (2\times60000)\right] \times 1\times10^{-6}} \checkmark\checkmark$$

$$= 6{,}871 \text{ Hz} \checkmark$$ (3)

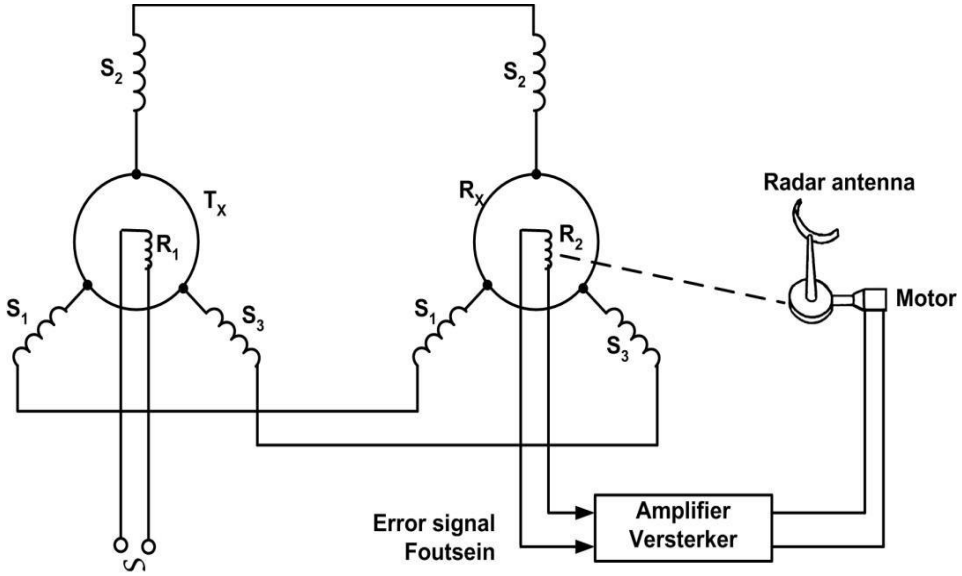
8.4
- Power supply (Change in DC)
- Output load (Change in output load)
- Temperature variations
- Operating point

(Any correct TWO) (2)

[12]

QUESTION 9: TRANSDUCERS

9.1 9.1.1

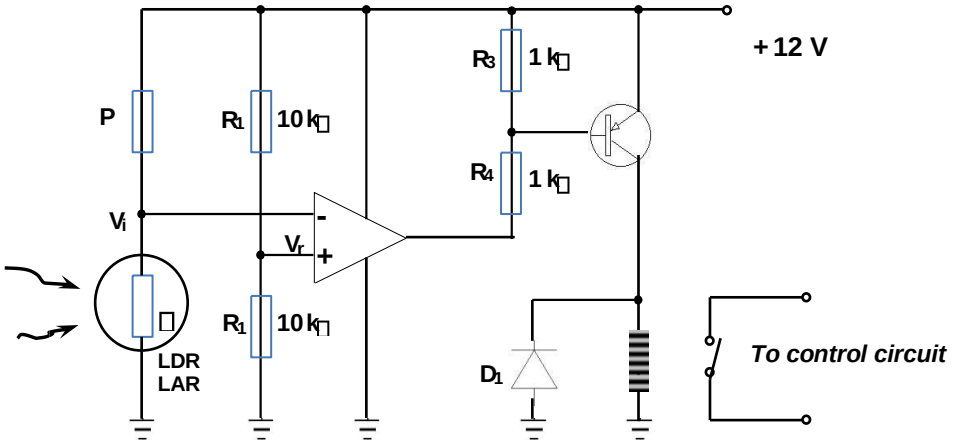


(5)

9.1.2

- The transmitter (Tx) and receiver (Rx) are in balance as soon as the rotors are at the same angle.✓✓
- If the rotor of the transmitter turns at an angle θ, the rotor of the receiver will produce an error signal to turn the motor.✓
- The antenna also turns the rotor of the receiver at the same angle θ so that the error signal becomes zero.✓ (4)

9.2



(One mark each for the correct component and label)
(Ignore the shading on resistors on the marking guideline. Students to be penalised for shading resistors.) (5)

[14]

TOTAL: 100

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